

The dimension dependence of local grid coloring

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Abstract

We study proper vertex coloring of full d -dimensional boxes and standard tori, treating the dimension as a variable. For every fixed palette $c \geq 4$, a linear dependence on d is necessary even for non-signaling outcomes with constant global success probability, and hence also for distributed quantum algorithms with input-independent pre-shared entanglement. Conversely, four colors suffice in $2^{O(d)}(1 + \log^* n)$ deterministic communication rounds, and in $2^{O(d)}$ non-signaling locality with exact success or quantum rounds with high probability. The upper bounds require no supplied coordinate directions. Classical deterministic and private-randomness algorithms also have the usual $\Omega_c(\log^* n)$ lower bound, with constants uniform in dimension.

The lower bound uses a random cyclic Cayley graph with high chromatic number and induced neighborhoods identical to grid neighborhoods. The upper bound combines nets at successively doubled scales with separated box boundaries. We prove the graph construction, the constant-success non-signaling transfer, and the geometric algorithm in full. Published non-signaling and quantum symmetry-breaking primitives are stated explicitly at their use. The remaining gap is linear versus single-exponential dependence on dimension.

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1 Results and conventions

For integers $d \geq 1$ and $L \geq 1$, the inputs are the full box $\{0, \dots, L-1\}^d$ and the standard torus $(\mathbb{Z}/L\mathbb{Z})^d$, with nearest-neighbor edges. Graphs are simple, including for small L . Write $n = L^d$. The maximum degree is at most $2d$ and, when $L \geq 3$, the torus is $2d$ -regular. A proper c -coloring assigns a member of $[c] = \{1, \dots, c\}$ to every vertex and different colors to adjacent vertices. The palette c is fixed independently of d and n in the asymptotic comparisons below.

Theorem 1.1 (Dimension-dependent bounds). *Fix $c \geq 4$. On the above grids with*

$$L \geq \max\{128, d^2\}, \quad (1)$$

the bounds in Table 1 hold, uniformly in d and L . All upper bounds use only four colors and hold for every $L \geq 1$. The dimension lower bounds hold even when globally consistent signed coordinate directions are supplied; the upper bounds also work without these directions. The lower bounds apply to boxes, standard tori, and tori promised to have even side length.

Model	Lower bound	Upper bound	Upper success
Deterministic LOCAL	$\Omega_c(d + \log^* n)$	$2^{O(d)}(1 + \log^* n)$	Certain
Private-randomness LOCAL	$\Omega_c(d + \log^* n)$	$2^{O(d)}(1 + \log^* n)$	Certain
Quantum LOCAL	$\Omega_c(d)$	$2^{O(d)}$	$1 - n^{-a}$
Non-signaling LOCAL	$\Omega_c(d)$	$2^{O(d)}$	Certain
Bounded-dependence LOCAL	$\Omega_c(d)$	$2^{O(d)}$	Certain

Table 1: Bounds for a fixed palette $c \geq 4$. Randomized-model lower bounds require global success at least $2/3$; deterministic lower bounds have a harmless additive constant. The quantum upper bound holds for each fixed requested $a > 0$. The private-randomness upper bound is the deterministic algorithm.

Here and throughout, success concerns the *entire* coloring, not just a fixed vertex or edge. The dimension lower bound also holds for any fixed positive global success probability, and allows shared randomness independent of the input topology. The additional randomized $\log^* n$ lower bound is stated for independent private randomness. The factor multiplying $\log^* n$ in the deterministic upper bound depends exponentially on d ; an additive $2^{O(d)} + O(\log^* n)$ bound is not proved here.

1.1 What the quantitative arguments add

Four-coloring grids in $O_d(\log^* n)$ deterministic rounds is known from Brandt et al. [2]; the underlying separated-box method is closely related to Holroyd–Schramm–Wilson [3]. Our upper-bound argument keeps the dependence on d explicit. Its key step is to construct a covering net using a sequence of auxiliary graphs of degree $2^{O(d)}$, instead of asking for a maximal independent set directly in a grid power of degree $2^{\Theta(d^2)}$. The separated-box conversion is then reproduced and proved.

Random-Cayley spectral methods are classical; see Alon [1]. The lower-bound construction combines their chromatic obstruction with the exclusion of short additive relations, so that every small induced neighborhood is a signed grid ball. The transfer to non-signaling outcomes uses the cheating-graph method of Coiteux-Roy et al. [6]. We give the complete transfer, including the amplification to constant global success and the finite size bookkeeping. The size-dependent classical lower bounds are due to Linial [4] and Naor [5]; the fast initialization used for the stronger-model upper bounds is due to Akbari et al. [7] and Flin–Li–Suomela [8].

The two main quantities to compare are thus $\Omega_c(d)$ and $2^{O(d)}$. The results concern constant palettes and full grid geometry. Rectangles with arbitrary aspect ratios, grids with holes, and arbitrary subgraphs are outside the geometric upper-bound theorem.

1.2 Inputs, distances, and computation

Vertices know d, L, n and whether the promised input is a box or a torus. They have distinct identifiers in $[n^C]$ for a fixed known constant $C \geq 1$; all assignments are permitted. Lower bounds use only identifiers in $[n]$. Local computation and message sizes are unrestricted. A communication round traverses one nearest-neighbor edge. Write dist_G for this graph distance and dist_∞ for maximum coordinate distance, using cyclic coordinate distances on a torus. In these graphs,

$$\text{dist}_G(u, v) \leq d \text{dist}_\infty(u, v).$$

Thus an ℓ_∞ radius r can require dr communication rounds. Constants in $O(\cdot)$ are independent of d, n unless a subscript indicates otherwise. The function $\log^*$ uses base two and stops on reaching a value at most one; $\log$ in explicit analytic inequalities is the natural logarithm.

A *frame* labels the incident directions consistently by $+e_i, -e_i$, $i \in [d]$. It supplies no absolute coordinates, origin, or parity bit. Lower bounds remain valid with this extra input. Absolute coordinates would instead make boxes and even tori zero-round two-colorable by coordinate-sum parity.

We use the usual radius convention: a vertex's T -round information is its rooted radius- T neighborhood with identifiers, degrees, ports, and visible input labels. Boundary stubs and their labels are included; data beyond them are not. Initially knowing neighbor identifiers changes the bounds by an additive constant. All constructions below leave an extra layer where a view must be embedded.

1.3 Non-signaling and quantum conventions

Fix a named processor set of size n and the public parameters. A *total outcome* assigns a probability distribution on simultaneous classical vertex outputs to every graph on this set, including graphs violating the grid promise. It has *non-signaling locality* T if, whenever two inputs have identical joint radius- T views of a set S , their output marginals on S agree. Joint views retain the actual processor names, identifiers, labels, and edges; they do not merely record separate isomorphism types. Correctness is required only on promised grid inputs. This is the total-outcome convention of [6].

Deterministic and randomized T -round algorithms induce such outcomes. So do T -round quantum algorithms: tracing out all systems outside the backward communication cones of S leaves a marginal determined by the operations and initial reduced state in those cones. Operations outside a cone cannot change that marginal. For the lower bounds, the initial state may be arbitrarily entangled and may depend on the processor names and public parameters, but must be independent of the unknown input topology. The processor set and state are kept fixed in the comparisons. The quantum upper bounds need no pre-shared entanglement; local register dimensions are unrestricted.

A bounded-dependence outcome, in addition to this cross-input locality, has independent output restrictions on sets whose graph distance exceeds a specified dependence range k . Its locality parameter here bounds both the non-signaling radius and k , up to absolute constant factors. This is a family of finite-input distributions. A finitely dependent process given only on $\mathbb{Z}^d$ need not extend to such a family, so the lower bounds below make no assertion about the dependence range of a standalone infinite-lattice process.

2 The linear dimension lower bound

We first construct a graph that looks like a grid throughout the region visible to an algorithm, yet cannot be colored with its palette.

Lemma 2.1 (A quantitative grid-like obstruction). *Let $d \geq 1$, $q \geq 2$, and $T \geq 0$ be integers. Put*

$$s = 2T + 3, \quad K = \binom{2d + s}{s}.$$

If

$$\log(64K) < \frac{d}{2(q-1)^2}, \quad (2)$$

there are a prime P with $8K < P < 16K$ and a simple $2d$ -regular graph H on P vertices with $\chi(H) > q$. Every induced closed radius- $(T+1)$ ball of H is isomorphic to the corresponding ball in $\mathbb{Z}^d$, preserving the individual signed directions.

Proof. Encode an integer vector by its positive and negative parts. Stars and bars then gives

$$\#\{z \in \mathbb{Z}^d : \|z\|_1 \leq s\} \leq \#\{w \in \mathbb{Z}_{\geq 0}^{2d} : \sum_i w_i \leq s\} = K. \quad (3)$$

Bertrand's postulate supplies a prime P strictly between $8K$ and $16K$. In particular $P > s$, since $K \geq s + 1$. Choose $a_1, \dots, a_d$ independently and uniformly from $\mathbb{Z}/P\mathbb{Z}$. For every nonzero z with $\|z\|_1 \leq s$, some coefficient z_i is nonzero modulo P . Conditioning on all other generators shows

$$\mathbb{P}\left[\sum_i z_i a_i = 0\right] = \frac{1}{P}.$$

The probability that any such relation holds is at most $(K-1)/P < 1/8$. Hence, except on an event of probability less than $1/8$, we have

$$\sum_i z_i a_i \neq 0 \quad \text{for } 0 < \|z\|_1 \leq s. \quad (4)$$

The operator that sums all shifts $\pm a_i$ has Fourier eigenvalues

$$\lambda_k = 2 \sum_{i=1}^d \cos(2\pi k a_i / P), \quad k \in \mathbb{Z}/P\mathbb{Z}; \quad \lambda_0 = 2d.$$

Indeed the characters $x \mapsto \exp(2\pi i k x / P)$ form an orthogonal basis and are eigenvectors of each shift. For fixed $k \neq 0$, the summands in λ_k are independent, have mean zero, and lie in $[-2, 2]$. Here is the needed concentration estimate. If X has these bounds and mean zero, convexity on $[-2, 2]$ gives

$$\mathbb{E} e^{-tX} \leq \cosh(2t) \leq e^{2t^2} \quad (t \geq 0).$$

The second inequality follows by comparing the power series, using $(2j)! \geq 2^j j!$. For a sum of d such independent variables, Markov's inequality and $t = u/(4d)$, for $u \geq 0$, therefore give

$$\mathbb{P}[\lambda_k \leq -u] \leq \inf_{t \geq 0} e^{-tu + 2dt^2} = e^{-u^2/(8d)}.$$

Taking $u = 2d/(q-1)$ and summing over the nonzero frequencies bounds the probability of any violation of

$$\lambda_k > -\frac{2d}{q-1} \quad (k \neq 0) \quad (5)$$

by

$$(P-1)e^{-d/(2(q-1)^2)} < \frac{16K}{64K} = \frac{1}{4}.$$

There is consequently a choice of generators satisfying both (4) and (5).

For this choice let

$$H = \text{Cay}(\mathbb{Z}/P\mathbb{Z}, \{\pm a_1, \dots, \pm a_d\}).$$

Since $s \geq 3$, the excluded relations include $a_i = 0$, $2a_i = 0$, and $a_i = \pm a_j$ for $i \neq j$. Thus the signed generators are nonzero and distinct: H is simple and $2d$ -regular, and the operator above is its adjacency matrix A .

We spell out the chromatic implication. Let S be a nonempty independent set of density $\alpha = |S|/P$. Since the graph has positive degree, $0 < \alpha < 1$. Set $x = 1_S - \alpha \mathbf{1}$. Then $x \perp \mathbf{1}$,

$$\|x\|^2 = P\alpha(1-\alpha), \quad x^\top Ax = -2dP\alpha^2.$$

Expanding in the nonconstant Fourier eigenvectors and using (5) yields

$$-2dP\alpha^2 > -\frac{2d}{q-1}P\alpha(1-\alpha), \quad \text{hence } \alpha < \frac{1}{q}.$$

Every color class is independent, so a q -coloring is impossible.

Finally define $\phi(z) = \sum_i z_i a_i$. Every walk in H lifts to a sequence of signed coordinate steps, so ϕ maps the lattice radius- $(T+1)$ ball onto the corresponding ball at zero in H . Two distinct points of that lattice ball cannot have the same image: their difference has norm at most $2T+2$. If their images are adjacent, subtracting the corresponding signed basis vector gives a relation of norm at most $2T+3$. By (4), this relation must be the zero vector, so that edge is exactly the corresponding signed grid edge. This proves both induced adjacency and preservation of each direction. Translation proves the assertion at every vertex of H . $\square$

2.1 Transfer without independence of the outputs

Lemma 2.2 (Constant-success amplification). *Suppose H has h vertices, $\chi(H) > q$, and all its induced radius- $(T+1)$ balls are signed grid balls. Let $N \geq 1$ and suppose*

$$Nh \leq L^d, \quad N \leq \left\lfloor \frac{L}{2T+5} \right\rfloor^d. \quad (6)$$

For every total non-signaling outcome of locality T , some legal named, ID-labeled grid with $n = L^d$ vertices has global q -coloring success probability at most

$$(1 - 1/h)^N \leq e^{-N/h}.$$

This holds for boxes and standard tori, including even tori when the prescribed L is even.

Proof. On the fixed named set $[n]$, form N disjoint copies of H and $n - Nh$ isolated vertices. Keep the same public values of d, L, n , grid-type promise, and fixed distinct identifiers as in the legitimate inputs. The outcome is defined on this graph even though correctness is not required there. Replace invalid output symbols by a fixed member of $[q]$; this preserves locality and cannot turn a successful legal coloring into a failure.

Call a vertex star bad if its center has the same color as one of its neighbors. Every realized output vector has at least one bad star in each copy of H . Independently of this vector, an analyst chooses a uniformly random center in each copy, with independent choices between copies. Conditional on any fixed vector, the chance that every selected star is good is at most $(1 - 1/h)^N$. Averaging over the vector and then over the choices implies that some *fixed*

selection of N centers has joint probability at most $(1 - 1/h)^N$ of all its star checks passing. No independence of algorithmic outputs has been used.

Let S be the union of these selected stars. In each copy of H , the radius- T neighborhood of its selected star is the radius- $(T + 1)$ ball at the center. Put $b = 2T + 5$ and partition a legal grid into $\lfloor L/b \rfloor^d$ disjoint coordinate blocks of side b , ignoring leftover strips. Embed one selected ball into each of N blocks, centered at coordinate $T + 2$ within the block. Its coordinate support lies between 1 and $b - 2$. Consequently these embedded balls have no edges between different blocks and lie away from a box boundary or a torus seam. They have the full degrees and signed ports needed for the views.

Preserve the actual names and identifiers in each embedded ball, and assign the unused names and identifiers bijectively to the remaining grid positions. There are enough vertices by (6). The joint radius- T view of S , including boundary degrees and ports, is identical on the two inputs. Non-signaling locality therefore equates their output marginals on S . A globally proper coloring of the genuine grid must pass all the selected star checks, proving the claim. $\square$

Theorem 2.3 (Uniform dimension lower bound). *Let $q \geq 2$ and assume (1). If a total non-signaling outcome of locality T properly q -colors every legal grid with global probability at least $2/3$, then*

$$T + 2 > \frac{d}{512(q - 1)^2 \log(q + 1)}. \quad (7)$$

For four colors one also has

$$T + 2 > d/256 \quad (d \geq 1024). \quad (8)$$

Both statements permit a supplied frame and a bipartiteness promise.

Proof. Put $a = (q - 1)^2$ and $b = \log(q + 1)$, and suppose $T + 2 \leq d/(512ab)$. Since $T \geq 0$, necessarily $d \geq 1024ab$. Also $s = 2T + 3 < d/(256ab) \leq d$. The elementary binomial estimate $\binom{m}{s} \leq (em/s)^s$ gives

$$\log K \leq s \log \frac{e(2d + s)}{s} \leq s \log \frac{3ed}{s}.$$

The function $x \log(3e/x)$ increases for $0 < x \leq 1$; hence

$$\log K \leq \frac{d}{256ab} \log(768eab) \leq \frac{10d}{256a}.$$

For the last step, use $b \geq \log 3$, $\log(768e) \leq 7b$, $\log a \leq 2b$, and $\log b \leq b$. Moreover $\log 64 \leq d/(256a)$ because $d/(256a) \geq 4b > \log 64$. Thus

$$\log(64K) \leq \frac{11d}{256a} < \frac{d}{2a},$$

and Lemma 2.1 applies. As $s \leq d$,

$$K \leq \binom{3d}{d} \leq 2^{3d} = 8^d, \quad h = P < 16 \cdot 8^d.$$

Take $N = 2h$. In this nonvacuous regime $d > 1024$ and $2T + 5 \leq d$. Consequently

$$Nh = 2h^2 < 512 \cdot 64^d \leq d^{2d} \leq L^d, \quad N < 32 \cdot 8^d \leq d^d \leq \left\lfloor \frac{L}{2T + 5} \right\rfloor^d.$$

For example, the first elementary inequality follows from $(d^2/64)^d \geq 16^d > 512$ for $d \geq 1024$; the second follows from $(d/8)^d \geq 128^d > 32$. Lemma 2.2 now bounds success by $e^{-2} < 2/3$, a contradiction.

For (8), suppose $T + 2 \leq d/256$ and $d \geq 1024$. Then $s < d/128$, and monotonicity of $x \log(e(2d + x)/x)$ gives

$$\log(64K) \leq \log 64 + \frac{d}{128} \log(257e) < \frac{d}{18}.$$

To check the last strict inequality without numerical approximation, use $\log 64 < 104/25$ and $\log(257e) < 131/20$. The corresponding margin is exactly

$$\frac{1}{18} - \frac{104/25}{1024} - \frac{131/20}{128} = \frac{37}{115200} > 0.$$

The logarithm bounds follow from the finite rational inequalities $\sum_{j=0}^{12} (104/25)^j / j! > 64$ and $\sum_{j=0}^{14} (111/20)^j / j! > 257$, and the exponential series. This is (2) for $q = 4$. The same estimates $s \leq d$, $2T + 5 \leq d$, and the same packing with $N = 2h$ apply. $\square$

For any fixed $\rho \in (0, 1]$, replacing $N = 2h$ by $N = \lceil h \log(2/\rho) \rceil$ makes the success upper bound at most $\rho/2$. The constant factor depending on ρ is absorbed by the packing inequalities for sufficiently large d . This proves $\Omega_q(d)$ for every fixed positive success probability in all the finite-input models of Table 1. The proof still uses only ordinary chromatic number: every run outputs one simultaneous classical color at every vertex.

3 Four colors with single-exponential dimension dependence

We first describe a deterministic algorithm with identifiers and a supplied frame. Later we remove the frame and replace the identifiers used for symmetry breaking by a locally proper initial coloring.

The only general-graph subroutine needed is the following elementary form of color reduction. A proof, including a uniform finite-table construction, is in Appendix A.

Lemma 3.1 (Color reduction and greedy independence). *On a graph of maximum degree at most a known $D \geq 1$, a supplied proper coloring with labels in $[m]$ can be reduced deterministically to at most $4096(D + 1)^6$ colors in $O(1 + \log^* m)$ rounds. A maximal independent set can consequently be computed in $O((D + 1)^6 + 1 + \log^* m)$ rounds. The constants are absolute; identifiers are unnecessary once the initial proper coloring is given.*

The polynomial exponent in this supporting lemma is unimportant for our dimension bound. Its useful features are uniform constants and the explicit dependence on the initial palette.

3.1 Sparse centers from successive scales

A set of grid vertices is *a-separated* if any two distinct members have maximum-norm distance greater than a . It *R-dominates* if every vertex has maximum-norm distance at most R from the set. All boxes in this section are maximum-norm balls intersected with the vertex set of the input graph.

Lemma 3.2 (Packing). *If S is a-separated, $a > 0$, then any maximum-norm ball of radius R contains at most $(1 + 2R/a)^d$ points of S . For a torus we assume $2R + a < L$.*

Proof. Put a continuous cube of side a around each selected point. Their interiors are disjoint, and all these cubes lie in a cube of side $2R + a$. The volume ratio gives the bound. For a box, the continuous cubes may extend past the input boundary, which does not affect this comparison. For a torus, lift the radius- R ball to coordinate intervals in $\mathbb{R}^d$; the enlarged cube has side less than L , and the same disjoint-volume comparison applies. $\square$

Lemma 3.3 (A net with covering radius below twice its spacing). *Let $s \geq 2$ be a power of two and, in the torus case, suppose $L > 40s$. An s -separated set that $(2s - 1)$ -dominates can be computed in*

$$O(ds(5^{6d} + 1 + \log^* n))$$

deterministic rounds.

Proof. Set $S_{-1} = V(G)$. For $j = 0, \dots, \log_2 s$ put $r_j = 2^j$, and form an auxiliary graph H_j on S_{j-1} by joining centers whose maximum-norm distance is at most r_j . Let S_j be a maximal independent set of H_j .

When $j = 0$ the closed neighborhoods have size at most 3^d . For $j \geq 1$, the set S_{j-1} is $r_j/2$ -separated, so packing bounds each closed neighborhood by $(1 + 2r_j/(r_j/2))^d = 5^d$. All these uses on a torus satisfy $2r_j + r_j/2 < L$. Thus each H_j has degree at most 5^d . Identifiers give an initial proper coloring with $m = n^C$ labels; Lemma 3.1 computes its maximal independent set in $O(5^{6d} + 1 + \log^* n)$ auxiliary rounds. Auxiliary neighbors can communicate in $O(dr_j)$ physical rounds. Routes and neighbors are determined by gathering the corresponding grid neighborhood; unbounded message sizes allow all routes to be used concurrently.

Maximality ensures that every point of S_{j-1} is within r_j of S_j . Induction gives separation r_j and domination radius at most $\sum_{i=0}^j 2^i = 2^{j+1} - 1$. Finally, $\sum_{j=0}^{\log_2 s} r_j < 2s$, proving the time bound and the claimed properties of the last set. $\square$

The covering radius $2s - 1$ is sufficient; we do not need an exactly maximal s -separated set. This is what permits the use of sparse auxiliary graphs throughout the construction.

3.2 Choosing boxes with separated faces

Fix

$$s = 2^{\lceil \log_2(16d25^d) \rceil}, \quad D_0 = 25^d, \quad R = 8s + 2. \quad (9)$$

In particular $16d25^d \leq s < 32d25^d$, so $s = 2^{O(d)}$. If $L \leq 40s$, gather the entire graph in $O(dL) = 2^{O(d)}$ rounds and choose a canonical proper three-coloring using the identifier order. Such a coloring exists: each coordinate path or cycle has a proper three-coloring, and the sum of its coordinate colors modulo three is proper on their Cartesian product. A single-vertex factor causes no difficulty. Exhaustive local search after gathering is allowed. This branch needs no supplied frame.

Henceforth suppose $L > 40s$. Obtain centers S from Lemma 3.3 and form the conflict graph J on S , joining centers at maximum-norm distance at most R . Packing gives

$$\Delta(J) + 1 \leq (1 + 2R/s)^d = (17 + 4/s)^d \leq 19^d \leq D_0.$$

The torus condition holds because $2R + s = 17s + 4 < L$. Use Lemma 3.1 to color J with $O((D_0 + 1)^6)$ colors, and process these color classes in increasing order. Each center a selects an integer radius

$$r_a \in \{2s + 1, \dots, 4s\}. \quad (10)$$

For every previously processed conflict neighbor b , every coordinate i , and both signs $\epsilon, \eta \in \{-1, 1\}$, require

$$\text{dist}(a_i + \epsilon r_a, b_i + \eta r_b) \geq 2. \quad (11)$$

This distance is ordinary coordinate distance in a box and cyclic distance on a torus. In a box we impose the condition even on face coordinates outside the input box.

For a fixed b, i, ϵ, η , at most three candidate radii violate (11): the face coordinates would have difference $-1, 0$, or 1 . On a torus these are three congruences modulo L , each with at

most one solution in the candidate interval, whose length $2s$ is less than L . Each previous neighbor thus forbids at most $12d$ radii. Since

$$2s > 12dD_0,$$

a permissible radius always exists; choose the least one. Centers processed in the same class are not conflict neighbors and make their choices independently.

Let $B_a = B_\infty(a, r_a)$ and call $I_a = B_\infty(a, r_a - 1)$ its *interior*. This is the clipped metric interior just defined, not an extra boundary condition at the outer boundary of the input box. The interiors cover the grid: each vertex is within $2s - 1$ of a center, whereas $r_a - 1 \geq 2s$. If the expanded boxes $B_\infty(a, r_a + 1)$ and $B_\infty(b, r_b + 1)$ intersect, their centers have distance at most $8s + 2 = R$. Their radii therefore satisfy (11).

3.3 The boundary-parity conversion

A vertex v is on the i -th border of center a if

$$v \in B_a, \quad v_i = a_i - r_a \text{ or } v_i = a_i + r_a,$$

with coordinate equality modulo L on a torus. Define

$$b(v) = \#\{(a, i) : v \text{ is on the } i\text{-th border of } a\} \pmod{2}. \quad (12)$$

Input-boundary faces created solely by clipping do not contribute to this count.

Lemma 3.4 (Containment of monochromatic components). *Every connected component induced by a single value of b is contained in some I_a . It is bipartite and has weak graph diameter at most $8ds$, meaning that any two of its vertices have distance at most $8ds$ in the whole graph G .*

Proof. Consider an edge uv with $u \in I_a$ and $v \notin I_a$. Only one coordinate, say j , changes. Thus v lies on exactly the j -th border of a , and u lies on no border of a .

No other center b can contribute a j -border at u or v . If it did, its expanded box would meet that of a , and the corresponding two parallel faces would have coordinate distance at most one, contrary to (11). For a coordinate $i \neq j$, the border contributions of every other box are unchanged along uv . Indeed the i -coordinate is unchanged; if membership in such a face changed, membership in its box would change along coordinate j , forcing one endpoint onto a j -border of that box. We have just excluded this. Consequently the unreduced count in (12) increases by exactly one, so $b(u) \neq b(v)$.

Every vertex is in some I_a . A path of constant b beginning there cannot leave I_a , because its first exit would cross an edge of the preceding kind. This proves containment. Coordinate diameters within I_a are less than $2r_a \leq 8s$, so ambient graph distances are at most $8ds$. Finally the containing box is bipartite. On a torus it lifts to an ordinary lattice box, since its side length is less than $8s + 1 < L - 1$; no wrapping edge joins its opposite faces. Clipping by a box boundary also preserves bipartiteness. $\square$

After computing b , each vertex gathers the labeled radius- $(8ds + 1)$ neighborhood. It contains its entire b -component and its boundary, by the weak diameter bound. Local computation identifies that component, chooses its minimum-identifier vertex as root, and assigns the parity of distance to the root *inside the component*. The component is bipartite, so this gives a well-defined bipartition bit $p(v)$. Its intrinsic diameter need not be small: the entire component has already been gathered using ambient graph distance.

Output $(b(v), p(v)) \in \{0, 1\}^2$. If an edge changes b , its endpoints already differ. Otherwise the endpoints lie in one component and their bipartition bits differ. The output is a proper four-coloring.

The conflict-graph color reduction and radius selection cost

$$O(dR((D_0 + 1)^6 + 1 + \log^* n))$$

physical rounds. Computing the border bits requires gathering center radii within distance $4s$, at cost $O(ds)$, and the last component step costs $O(ds)$. Together with the net and small-side branches, an explicit sufficient bound is

$$O(d^2 25^{7d} + d^2 25^d (1 + \log^* n)) = 2^{O(d)} (1 + \log^* n). \quad (13)$$

This proves the deterministic upper bound with a frame.

3.4 Recovering the necessary geometry without a frame

The large-side branch has $L > 40s \geq 5$. In a full box or standard torus of side at least five, two incident edges belong to different coordinate axes if and only if they lie together in a four-cycle. Every such four-cycle is a coordinate square. Thus a radius-two view identifies the groups of incident edges belonging to one axis: opposite pairs and, at a box boundary, singletons.

A root may name these groups and signs arbitrarily. Across an edge, each transverse direction is transported through its unique coordinate square. The back edge receives the opposite sign, and the direction of travel is transported by straight continuation, when the continuation edge exists; a missing continuation records a boundary. In a neighborhood of graph radius t , radius $t + 2$ provides all the squares required for this transport. To justify consistency, compare the procedure with the actual Cartesian-product representation of the promised input: at the root they differ by a signed permutation of axes, and every square and straight-continuation rule preserves that agreement. Coordinates are taken modulo the known L on tori. The accumulated displacement of any path therefore gives the correct relative coordinates, including when the neighborhood wraps around the torus.

All geometric tests used above are invariant under signed permutations: maximum-norm distance, cubes, parallel-face separation, and the total border count. The chosen centers depend on auxiliary colors, the radius choice is the least feasible integer, and the component root depends on identifiers. Different processors need not use the same axis names. An additional two layers suffice for each geometric gathering; the sum of these costs is already covered by (13). This proves the unframed upper bound.

4 Removing the size term in the stronger models

The geometric construction needs symmetry-breaking labels that distinguish nearby vertices, rather than globally unique labels. We now state precisely the published ingredients used to obtain such labels. These are supporting general-graph results; all dependence on grid dimension is tracked in the subsequent reduction.

Theorem 4.1 (Cited initialization primitives). *For finite graphs of maximum degree at most a known $D \geq 1$:*

1. *There is an exact proper 3^D -coloring distribution with dependence range and non-signaling locality $O(1 + \log^* D)$. The family is consistent under subgraph isomorphisms of enlarged neighborhoods, in particular under equal named local views on different inputs.*
2. *With n and D known, a quantum algorithm starting from a product state produces a proper $O(D^2)$ -coloring in $O(1 + \log^* D)$ rounds with global probability at least $1 - n^{-a}$ for every fixed requested $a > 0$.*

The constants in the dependence on D are uniform.

Part 1 is the intermediate 3^D -coloring in the *proof* of Lemma 6.14 of Akbari et al. [7], obtained from their rooted-pseudoforest coloring (Lemma 6.11) and composition lemmas (6.4 and 6.8). We use their corrected February 2026 version. The statement of their Lemma 6.14 additionally reduces the palette to $D + 1$ at a larger cost; that further step is not used here. Part 2 is Theorem 6.5 of the September 2026 preprint of Flin–Li–Suomela [8]. Its local quantum resources can depend on n and the accuracy parameter. It does not claim exact success. These are the two imported symmetry-breaking theorems in the stronger-model upper bounds.

Lemma 4.2 (Power graphs and postprocessing). *Let G^r join distinct vertices at G -distance at most r , where $r \geq 1$ is an integer. A total non-signaling outcome of locality t on G^r pulls back to locality $O(r(t + 1))$ on G . A dependence range k becomes at most rk . Deterministic b -round postprocessing adds b to non-signaling locality and at most $2b$ to dependence range. A quantum round on G^r can be simulated in $O(r)$ rounds on G , after $O(r)$ setup.*

Proof. The radius- t joint view of a set in G^r is determined by its radius- $r(t + 1)$ view in G . The extra r determines all power edges incident to visible vertices, including the short paths witnessing those edges. Equality of these original views therefore implies equality of the required power-graph views and output marginals. If two sets have G -distance greater than rk , no path of at most k power edges can join them; the dependence assertion follows.

A b -round postprocessing reads only the b -neighborhood of its output set. The neighborhoods of two sets at distance greater than $rk + 2b$ remain separated by more than rk , which proves the composition bounds. For quantum simulation, gather the physical radius- r neighborhoods and fix a route of length at most r for each ordered pair of virtual neighbors, using identifiers to break ties. Route each virtual message along its path. All registers may travel concurrently because capacities are unrestricted. A quantum register is forwarded along its route, never copied. $\square$

Theorem 4.3 (Four colors with no size-dependent time term). *Full balanced boxes and standard tori admit exact four-coloring outcomes in NS-LOCAL and BD-LOCAL with locality $2^{O(d)}$. They admit $2^{O(d)}$ -round quantum four-coloring algorithms with global probability at least $1 - n^{-a}$ for any fixed $a > 0$, without pre-shared entanglement. No frame is required.*

Proof. Use s, R from (9) and set

$$r = 64ds + 2, \quad D_{\text{pow}} = (2d)^{r+1}. \quad (14)$$

The latter is an upper bound on the degree of G^r whenever G has maximum degree at most $2d$: the number of reachable vertices is at most $\sum_{j=0}^r (2d)^j < (2d)^{r+1}$. Every auxiliary edge used by the geometric construction has physical length at most $dR < r$. Every final monochromatic component has weak diameter at most $8ds < r$.

For non-signaling and bounded dependence, apply part 1 of Theorem 4.1 to G^r . Let X denote the resulting proper coloring, with labels in $[Q]$, $Q = 3^{D_{\text{pow}}}$. By Lemma 4.2, its locality and dependence range on G are $O(r(1 + \log^* D_{\text{pow}}))$.

Run the geometric algorithm using X as the initial coloring in every invocation of Lemma 3.1. This is valid even though the sets of centers are chosen adaptively: all edges of all auxiliary graphs have length at most r , so restriction of X to any set of retained centers is still a proper coloring of the relevant auxiliary graph. Its palette bound remains Q ; the size term in the algorithm is therefore $1 + \log^* Q$ instead of $1 + \log^* n$. For component bipartitions, the minimum X -label is unique within each component because its weak diameter is less than r . Identifiers may also be retained for gathering and root choice; their magnitude never enters the remaining round bound.

If $L \leq 40s$, the whole grid has diameter at most $dL < r$, so its X -labels are distinct and the whole-graph branch works within the same bound. Otherwise use the separated-box construction. Quantitatively,

$$r = 2^{O(d)}, \quad D_{\text{pow}} \leq 2^{2^{O(d)}}, \quad \log^* Q = O(1 + \log^* d).$$

The initialization and the deterministic postprocessing each take $2^{O(d)}$ locality. Lemma 4.2 proves the same bound for dependence range. Correctness is exact.

For quantum computation, first handle $n = 1$ directly and otherwise apply part 2 of Theorem 4.1 once to the entire power graph, with degree bound $\min\{D_{\text{pow}}, n - 1\}$. This gives $Q = O(D_{\text{pow}}^2)$ initial labels, so again $\log^* Q = O(1 + \log^* d)$. Simulating the virtual rounds by Lemma 4.2 costs $2^{O(d)}$ physical rounds. Conditioned on the one event that this initial coloring is proper, every later step is deterministic and correct. Thus the original global success guarantee $1 - n^{-a}$ is preserved; no union bound over the later auxiliary computations is needed. $\square$

For completeness, these constructions have total outcomes on inputs violating the promise. First delete every edge incident to a vertex of degree greater than $2d$, a local operation that changes no legal grid. Form the power graph of the resulting bounded-degree graph and use the total initialization family. Run the geometric postprocessing for its fixed prescribed time; when a local view fails a required geometric test, use a fixed fallback rule. On all inputs this is a bounded-radius deterministic postprocessing, whether or not its coloring is proper. Hence the resulting family satisfies exactly the cross-input locality convention used in the lower bound.

5 The classical size term, uniformly in dimension

For fixed dimension the classical size dependence is familiar. We include the reduction that keeps its constants independent of d , and a short proof of the deterministic ingredient.

5.1 A deterministic path and cycle lower bound

For $m \geq 1$, let $S_m(L)$ have as vertices the strictly increasing m -tuples from $[L]$. Join the two consecutive m -windows of each increasing $(m + 1)$ -tuple. In particular $S_1(L) = K_L$. If f is a k -coloring of $S_m(L)$ for $m \geq 2$, assign to $(x_1, \dots, x_{m-1})$ the subset

$$\{f(x_1, \dots, x_{m-1}, y) : y > x_{m-1}\} \subseteq [k].$$

This is a proper coloring of $S_{m-1}(L)$ with at most 2^k colors. Indeed, for an increasing m -tuple, the color $f(x_1, \dots, x_m)$ belongs to the subset for its prefix and does not belong to the subset for its suffix: every extension of the suffix is adjacent to the original tuple. Empty subsets cause no exception to this argument. Iterating yields

$$L \leq \underbrace{2^{2^{\cdot^{2^k}}}}_{m-1 \text{ exponentiations}} \quad \text{whenever } S_m(L) \text{ is } k\text{-colorable.} \quad (15)$$

Now fix all public parameters of a deterministic T -round algorithm on an oriented L -cycle, with identifiers in $[L]$. If $L \geq 2T + 4$, its output on an increasing local ID sequence defines a function on $S_{2T+1}(L)$. Each pair of adjacent windows can be embedded in one cycle ID assignment, completing the unused identifiers arbitrarily. Correctness forces the two outputs to differ. A c -color algorithm therefore induces a c -coloring of this shift graph. By (15),

$$2T \geq \log^* L - \log^* c.$$

For an oriented path, place the windows in the interior, leaving one unused vertex at either end; the same safe condition $L \geq 2T + 4$ suffices. If that condition fails, T is already linear in L , which implies the stated iterated-logarithmic lower bound up to an absolute additive constant. Restricting to even L does not change the proof. This is the standard shift-graph argument underlying Linial's lower bound [4].

5.2 Private randomness

We use the following precise published ingredient: for each fixed $\rho > 1/2$, private-randomness three-coloring of directed L -cycles, with distinct adversarial identifiers in $[L]$ and global success at least ρ , requires $\Omega(\log^* L)$ rounds [5, Theorem 2.1]. The proof allows arbitrary local decision functions for each fixed L ; knowledge of L and other fixed public parameters does not weaken the bound.

It extends to any fixed palette $c \geq 3$: after a proper c -coloring, process colors $c, c-1, \dots, 4$, replacing each by an available member of $\{1, 2, 3\}$. A cycle vertex has at most two neighbors, and the simultaneously processed vertices form an independent set. This takes $c-3$ rounds and preserves success whenever the initial coloring is proper.

Here is an explicit reduction covering paths as well. Run a putative T -round path algorithm on a cycle with the same public size, interpreting its interior local rule at every vertex. As in the dimension lower bound, normalize any output outside $[c]$ to a fixed color. For the analysis only, choose a uniformly random edge to cut, independently of all private tapes, and couple the path and cycle runs by using the same tapes and IDs. If the cycle output has a bad edge, select one deterministically from the realized vector. Except for at most $2T+3$ choices of cut edge, that bad edge and the views producing its outputs are unchanged on the resulting path. If every path succeeds with probability at least $2/3$, averaging over cuts gives

$$\mathbb{P}[\text{cycle failure}] \left(1 - \frac{2T+3}{L}\right) \leq \frac{1}{3}.$$

For $T \leq L/16$ and $L \geq 24$, the parenthesized factor is at least $3/4$, so cycle success is at least $5/9 > 1/2$ and Naor's bound applies. Otherwise T is already linear in L , or L is bounded. Thus paths have the same $\Omega_c(\log^* L)$ lower bound up to an additive constant. The cycle result and its proof also apply when only even lengths are considered.

5.3 Simulating a grid by a path or cycle

One processor of an oriented L -cycle can simulate an entire $(d-1)$ -dimensional cross-section of an L^d -vertex torus. Index the L^{d-1} transverse positions by $j \in \{0, \dots, L^{d-1} - 1\}$. At a cycle processor with ID $i \in [L]$, give its virtual vertex j the identifier

$$(i-1)L^{d-1} + j + 1 \in [n].$$

These are all distinct. Edges in transverse directions are internal to a simulator; edges in the first direction connect adjacent simulators. Unbounded messages permit one grid round to be simulated in one cycle round. The supplied grid frame, if any, is also simulated consistently. Reading the output at one fixed transverse position produces a proper c -coloring of the cycle whenever the grid output is proper.

For randomized algorithms, each simulator draws the collection of private tapes for all its virtual vertices. Collections at distinct simulators are independent, exactly as required by the private-randomness lower bound. For boxes use an oriented L -path and transverse boxes. The rule may depend on d , on $n = L^d$, and on the grid type; these are fixed public parameters in the one-dimensional lower bound. We have proved

$$T \geq a_c \log^* L - b_c$$

with constants independent of d .

Finally, under (1) we have $d \leq \sqrt{L}$ and $\log_2 L \leq \sqrt{L}$ for $L \geq 128$. Thus $\log_2 n = d \log_2 L \leq L$ and

$$\log^* n \leq 1 + \log^* L.$$

Combining the dimension and size lower bounds, using $\max\{x, y\} \geq (x + y)/2$, gives $\Omega_c(d + \log^* n)$ up to an additive constant for deterministic and private-randomness LOCAL. This finishes the lower-bound claims of Theorem 1.1.

6 The remaining dimension gap

For each fixed $c \geq 4$, even non-signaling outcomes must see a distance proportional to d , while the construction here uses a distance exponential in d . When L is of order d^2 , $\log^* n = \log^* d + O(1)$, so the lower bound is a genuine dimension obstruction rather than a disguised size lower bound. In particular it rules out $O(\log d + \log^* n)$ time for a fixed palette in the classical models and $O(\log d)$ locality in the stronger models.

There is a concrete limitation of the separated-box proof. Suppose radius assignment must work for every s -separated, s -dominating net, using the interval $\{2s + 1, \dots, 4s\}$ and condition (11) whenever expanded boxes intersect. Consider the 2^{d-1} centers

$$\{0, s + 1\}^{d-1} \times \{0\}.$$

They are s -separated and can be extended to a maximal such set in a sufficiently large finite box, which is then s -dominating. Every pair of their boxes overlaps because their center distance is $s + 1$, while their radii exceed $2s$. All their positive faces in the last coordinate are at positions r_a . Condition (11) therefore requires these 2^{d-1} radii to be pairwise at least two apart. The interval of $2s$ consecutive candidate integers accommodates at most s such radii. Necessarily $s \geq 2^{d-1}$.

Thus a radius rule that accommodates arbitrary nets in this way already needs an exponential scale. This does not prove an exponential coloring lower bound. A more structured net, a different covering, or a different conversion of a covering to a coloring could improve the upper bound. Whether the dimension dependence for four colors can be polynomial is not settled by these arguments.

A An elementary color-reduction subroutine

We prove Lemma 3.1 so that every deterministic step of the geometric upper bound is specified. The following standard set-system method only needs a polynomial number of colors in the degree; optimizing its exponent is unnecessary.

Fix integers $D \geq 1$ and $m \geq 2$. There are sets $A_1, \dots, A_m \subseteq [M]$, where

$$M \leq \left\lceil 8(D + 1)^2 \log(m + 1) \right\rceil, \tag{16}$$

such that for every i and every $J \subseteq [m] \setminus \{i\}$ with $|J| \leq D$,

$$A_i \setminus \bigcup_{j \in J} A_j \neq \emptyset. \tag{17}$$

To see this, put each universe element in each set independently with probability $p = 1/(D + 1)$. For a fixed pair (i, J) , a given element witnesses (17) with probability

$$p(1 - p)^{|J|} \geq \frac{1}{D + 1} \left(\frac{D}{D + 1} \right)^D \geq \frac{1}{4(D + 1)}.$$

The last inequality follows from $(1 + 1/D)^D < e < 4$. There are at most $(m + 1)^{D+1}$ pairs (i, J) : encode i and list the members of J in D padded positions. With $M = \lceil 8(D + 1)^2 \log(m + 1) \rceil$, a union bound bounds the probability that any pair has no witness by

$$(m + 1)^{D+1} \exp\left(-\frac{M}{4(D + 1)}\right) \leq (m + 1)^{-(D+1)} < 1.$$

Hence such a family exists. It is a deterministic, uniformly computable finite table: enumerate $M = 1, 2, \dots$ and all membership tables, stopping at the first that satisfies the finite tests (17). The argument proves that this search terminates within (16). Its computational cost consumes no communication rounds.

Given a proper $[m]$ -coloring of a degree- D graph, vertices exchange their colors once. A vertex of color i chooses the least element of A_i outside the sets of its neighbors' colors. This is possible by (17). If its neighbor has old color j , the chosen element is outside A_j , whereas that neighbor's new color belongs to A_j . Thus the new coloring is proper and uses at most M colors.

Let $b = 16(D + 1)^2 \geq 64$. For $m \geq 2$, the palette bound is

$$M \leq b \log_2 m.$$

Indeed $\log(m + 1) \leq 2 \log m$ and $\log 2 < 3/4$, so $M \leq 12(D + 1)^2 \log_2 m + 1 \leq 16(D + 1)^2 \log_2 m$. If $m \geq 2^{b^2}$, two applications reduce the palette to at most

$$b \log_2(b \log_2 m) \leq \log_2 m.$$

For the inequality, write $x = \log_2 m \geq b^2$; the function $x - b(\log_2 b + \log_2 x)$ is increasing there and at $x = b^2$ has value $b(b - 3 \log_2 b) \geq 0$. Every pair of reductions therefore removes at least one iterated-logarithm level until the palette is at most 2^{b^2} . One further reduction gives at most $b^3 = 4096(D + 1)^6$ colors. An already smaller palette can simply be kept. This uses $O(1 + \log^* m)$ rounds, with an absolute constant. A proper one-color input is edgeless and needs no reduction.

Finally process the reduced color classes in order. A vertex joins an independent set if none of its already processed neighbors has joined. Simultaneously processed vertices are nonadjacent. Every excluded vertex has a selected neighbor, so the resulting independent set is maximal. The sweep takes at most $4096(D + 1)^6$ rounds. This proves the lemma, including its uniformity and its dependence on the initial palette.

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